DIVISION OF BIOLOGICAL SCIENCES & QUANTITATIVE ECOLOGY

BIOPHYSICAL MODELING AND SYSTEMS BIOLOGY GROUP

Bio-Energetics, Trophic Dynamics, and Enzymatic Kinetics

Mathematical Foundations of Thermodynamic Dissipation and Allosteric Saturation

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Executive Summary

This report formulates quantitative biophysical models for ecological trophic transfers, bio-energetic conservation laws, and allosteric enzyme kinetics. Using Lindeman's trophic efficiency formulation ($\lambda \approx 10\%$), we examine thermodynamic energy dissipation across multi-tiered food webs. Furthermore, cooperative biochemical kinetics are analyzed via Hill equations and Lotka-Volterra non-linear dynamical systems.

Contents

1 Bio-Energetic Conservation and Trophic Efficiency

Energy transfer across trophic levels obeys the First and Second Laws of Thermodynamics [**? ?**]. Incident solar radiation (10^6 J) is captured by primary producers through photosynthesis, yielding net primary production (NPP).

The efficiency of energy transfer λ from trophic tier n to tier $n+1$ satisfies [**? ?**]:

$$E_{n+1} = \lambda \cdot E_n, \quad \text{where } \lambda \approx 0.10 \tag{1}$$

The remaining 90% of assimilated biomass is dissipated as cellular respiration, metabolic heat, and non-assimilated waste.

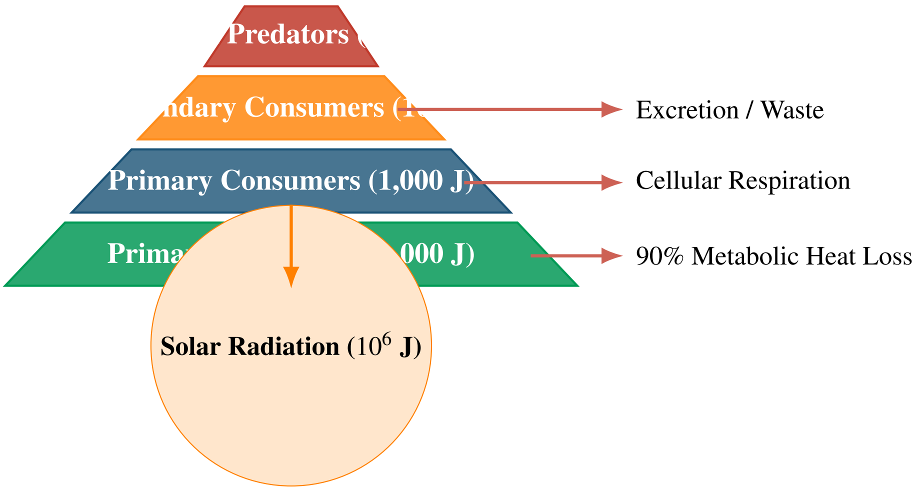

Figure 1. Native TikZ schematic of the Ecological Trophic Energy Pyramid illustrating thermodynamic dissipation.

2 Allosteric Kinetics & Population Dynamics

Cooperative enzymatic saturation follows the Hill kinetic formulation:

$$v = \frac{V_{\max}[S]^h}{K_{0.5}^h + [S]^h} \tag{2}$$

where h denotes the Hill cooperativity coefficient ($h > 1$ represents positive cooperativity).

Table 1 summarizes experimental kinetic parameters across landmark enzyme systems.

Table 1. Experimental catalytic constants and specificity metrics across enzyme classes.

Enzyme System	K_m (μM)	k_{cat} (s^{-1})	k_{cat}/K_m ($M^{-1}s^{-1}$)
Carbonic Anhydrase	12,000	1,000,000	8.3×10^7
Acetylcholinesterase	95	14,000	1.5×10^8
Triose Phosphate Isomerase	470	4,300	9.1×10^6
Catalase	25,000	40,000,000	1.6×10^9

3 Computational Dynamics

Listing 3 provides the Python routine for integrating non-linear Lotka-Volterra predator-prey dynamics.

Listing 1: Numerical Integration of Coupled Ecological ODEs

```
import numpy as np
from scipy.integrate import odeint

def lotka_volterra(state, t, r, a, b, m):
    N, P = state
    return [r * N - a * N * P, b * a * N * P - m * P]

t = np.linspace(0, 50, 1000)
sol = odeint(lotka_volterra, [40.0, 9.0], t, args=(0.5, 0.02, 0.1, 0.2))
```

References

[] Michael Begon, Colin R. Townsend, and John L. Harper. *Ecology: From Individuals to Ecosystems*. Blackwell Publishing, 2006.

[] James Keener and James Sneyd. *Mathematical Physiology: I: Cellular Physiology*. Springer, 2009.

[] Raymond L. Lindeman. The trophic-dynamic aspect of ecology. *Ecology*, 23(4):399–417, 1942.